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Started	Wednesday, 28 October 2026, 7:00 PM
Completed	Wednesday, 28 October 2026, 7:03 PM
Duration	3 mins 4 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. The list must be sorted

c. There should be direct access to the middle element in any sublist

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99

b. 90 and 100

c. 89 and 94

d. 90 and 99 ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎

b. O(n log n)

c. O(log n)

d. O(1) ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array

b. There must be a mechanism to access middle element directly

c. Binary search algorithm is not efficient when the data elements more than 1500 ✓

d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓

b. It does not work with negative numbers

c. It is slower than linear search for small lists

d. It can only be applied to large lists

Question **6**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓

b. When the list contains duplicate elements

c. When the search algorithm finishes

d. When the list is sorted

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element

b. By dividing the list length by two ✓

c. By using a hash function

d. By comparing each element sequentially

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **9**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search can find multiple instances of the target element
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search works on unsorted lists

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list is sorted ✓
c. When the list is very small
d. When the list contains duplicate elements

Question **11**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Interpolation search
c. Linear search ✓
d. Binary search

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(n log n)
c. O(1)
d. O👎 ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues in the left or right sublist ✓
b. The search stops
c. The search continues from the beginning
d. The list is sorted

Question **14**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. 1
c. log n
d. n ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(log n) ✓
c. O👎
d. O(n log n)